Lecture Plan

B.Tech. CSE-Cyber Security

Subject Name: Cloud Administration 30 hrs +10 experiments

Subject Code:

Sl. No.	Description	Bloom's level*
CO1	Explain cloud computing concepts, service models, deployment models, and AWS global infrastructure.	Understand
CO2	Deploy and manage compute and server-less services in cloud environments.	Apply
CO3	Implement cloud storage, backup strategies, and database services.	Apply
CO4	Configure networking components and identity management services in cloud infrastructure.	Analyze
CO5	Monitor cloud resources and implement governance and cost optimization strategies.	Analyze, Evaluate

S.No.	Topic	Lecture No.
1.	**Unit 1 – Cloud Computing and AWS Fundamentals: Introduction to Cloud Computing** • Cloud computing concepts • Service models (IaaS, PaaS, SaaS) • Deployment models (Public, Private, Hybrid Cloud)	**Lecture 1**
2.	**Cloud Infrastructure Fundamentals** • Virtualization fundamentals • Basic cloud architecture concepts • Benefits and challenges of cloud computing	**Lecture 2**
3.	**AWS Global Infrastructure** • AWS Regions • Availability Zones • Edge Locations • AWS Global Network	**Lecture 3**
4.	**Cloud Security Fundamentals** • AWS Shared Responsibility Model • Fundamental cloud security principles	**Lecture 4**
5.	**Case Studies and Cloud Deployment Models** • Real-world cloud deployments • Cloud architecture review	**Lecture 5**
6.	**Unit 2 – Compute and Serverless Services:Amazon EC2 Fundamentals** • EC2 overview	**Lecture 6**

	• Instance types • Amazon Machine Images (AMIs)	
7.	**EC2 Configuration** • Pricing models • Key pairs • Tenancy options • Launching EC2 instances	**Lecture 7**
8.	**Compute Scalability** • Auto Scaling fundamentals • Compute scalability best practices • High availability concepts	**Lecture 8**
9.	**Platform Services** • Elastic Beanstalk • Amazon Lightsail • Deployment strategies	**Lecture 9**
10.	**Serverless Computing** • AWS Lambda • Event-driven architecture • Serverless deployment • Monitoring • Serverless use cases and limitations	**Lecture 10**
11.	**Unit 3 – Storage and Backup Services:** **Amazon S3 Fundamentals** • S3 architecture • Storage classes • Lifecycle management	**Lecture 11**
12.	**Advanced S3 Features** • Versioning • Cross-region replication • Amazon S3 Glacier	**Lecture 12**
13.	**Elastic Block Storage** • Amazon EBS volume types • Snapshots • Volume management	**Lecture 13**
14.	**Shared Storage Services** • Amazon EFS • Amazon FSx family • Data durability and availability concepts	**Lecture 14**
15.	**Backup and Disaster Recovery** • AWS Backup services • Disaster recovery basics • **Unit Revision**	**Lecture 15**
16.	**Unit 4 – Database and Container Services:** **Amazon RDS** • Database engines • Multi-AZ deployment • Read Replicas	**Lecture 16**
17.	**NoSQL and Caching Services** • Amazon DynamoDB • Indexes	**Lecture 17**

	• DynamoDB Accelerator (DAX) • Amazon ElastiCache	
18.	**Database Design** • Database selection strategies • Workload optimization	**Lecture 18**
19.	**Container Services** • Containerization concepts • Amazon ECS • Amazon EKS • Amazon ECR	**Lecture 19**
20.	**Container Deployment** • Container orchestration • Container best practices • **Unit Revision**	**Lecture 20**
21.	**Unit 5 – Networking and Security Services:** **Amazon VPC** • VPC architecture • Subnets • Route Tables • Internet Gateway	**Lecture 21**
22.	**VPC Networking** • NAT Gateway • Security Groups • Network ACLs (NACLs) • VPC Peering • Transit Gateway	**Lecture 22**
23.	**Hybrid Connectivity** • AWS Direct Connect • VPN Connectivity • Route 53 DNS Services • Routing Policies	**Lecture 23**
24.	**Content Delivery and Identity Management** • CloudFront CDN • Global Accelerator fundamentals • IAM Users, Groups and Roles • IAM Policies • Permission Boundaries	**Lecture 24**
25.	**AWS Security Services** • AWS WAF • AWS Shield • Amazon GuardDuty • AWS Secrets Manager • AWS Certificate Manager • SSL/TLS Management	**Lecture 25**
26.	**Unit 6 – Monitoring, Integration and Cost Management:** **Application Integration Services** • Amazon SNS • Amazon SQS • Amazon EventBridge • Decoupled architecture	**Lecture 26**

	• Scalability design	
27.	**Monitoring and Logging** • Amazon CloudWatch • Logs • Metrics • Alarms	**Lecture 27**
28.	**Governance Services** • AWS CloudTrail auditing • AWS Config • AWS Trusted Advisor • AWS Systems Manager Automation	**Lecture 28**
29.	**Cost Management** • AWS Organizations • AWS Budgets • Cost Explorer • Cost and Usage Reports	**Lecture 29**
30.	**Cloud Cost Optimization and Course Revision** • Savings Plans • Reserved Instances • Cloud architecture decisions impacting cost optimization • Complete Course Revision	**Lecture 30**

List of Experiments

Experiment No.	Practical Experiment
Experiment 1	Create and configure an AWS account and IAM users with roles and policies.
Experiment 2	Launch and configure an Amazon EC2 instance and connect via SSH.
Experiment 3	Configure Auto Scaling and Elastic Load Balancer for high availability.
Experiment 4	Create and manage Amazon S3 buckets with lifecycle policies.
Experiment 5	Configure Amazon EBS volumes and snapshots for EC2 instances.
Experiment 6	Deploy a serverless application using AWS Lambda.
Experiment 7	Create and configure Amazon RDS database instances.
Experiment 8	Deploy a containerized application using Amazon ECS or EKS.
Experiment 9	Configure VPC, subnets, route tables, and internet gateway.
Experiment 10	Monitor cloud resources using Amazon CloudWatch metrics and logs.

Suggested Readings and References

Sr. No.	Author	Name of the Book	Publisher	Year of Publication
1	Mark Wilkins	AWS Certified Solutions Architect – Associate (SAA-C03) Cert Guide	Pearson	2023